

getting feedback on their mistakes and then iterating to improve them. A study by researchers at the Queensland University of Technology revealed that VR simulation improved the technical skills of undergraduate medical imaging students.

United Rentals, for instance, uses immersive learning for open space ratios to bring construction sites into the classroom and claims to have found a forty per cent reduction in training time. Walmart also uses VR to interview and train employees while Verizon has been using it to train its retail workers in handling robberies in stores.

Others

Digital learning guarantees more reach for the current generation of students. Apart from the above mentioned technologies, innovations like video-surveillance via Internet of Things (IoT) infrastructure and connected devices increase security, reliability, and capabilities for educational settings.

To utilise the huge amount of data we generate every day, data management software for data analytics makes the necessary data sets easy to understand. Enrollment reports, registration level statistics, and analysis of other such information can boost productivity and decrease costs.

Indian scenario

India has one of the largest higher education systems in the world. There have been numerous issues when it comes to the education sector employing technologies in the country, especially in rural areas. Many educational institutions still don't have a central repository that collects and organises data. A lot of these don't have standardised and digitalised things like certificates, marksheets, diplomas, and degrees. Other issues include outdated teaching methods, unavailability of well-trained teachers, inadequate student-teacher ratio, lack of enough teaching resources, and so on.

But with the Internet usage growing rapidly, the foundation for digital education is being laid. Take the example of SP Robotic Works, a company that provides an AI-assisted learning platform for teaching robotics and coding.

This, however, comes with a host of issues like poor connectivity and lack



Mixed reality powered smart glasses, the Jio Glass, launched by Reliance Jio in India (Credit: www.freepressjournal.in)

of proper knowledge about how to use these digital technologies and tools. Also, the content is not available in all the regional languages, which may be a barrier for many individuals. Sometimes there are budget constraints in the initial implementation of efficient hardware and software too. Moreover, maintenance is often neglected. Promising models that can help at a large scale need to be introduced.

There are many restrictions currently with the technologies themselves. For some situations it is yet difficult to program machines to understand complex human interactions that aid people in making better decisions.

However, with the government's focus on Digital India and digitisation of higher education, there are a growing number of innovations and startups working in this field. India being a leader in Information and Communication Technology (ICT) as well as other cutting-edge domains can provide an upper hand. Top institutes in the nation are working in collaboration with the HRD Ministry to develop a comprehensive structure and policy for implementing a robust online education system.

In May, union finance minister Nirmala Sitharaman launched the PM eVidya programme to promote digital education and make e-learning feasible for students and teachers. Under this, the e-learning portal DIKSHA provides e-content and QR coded energised textbooks for all grades, which will benefit nearly 250 million children in various states and union territories. The education content is currently available in fourteen languages, including Kannada, Malayalam, Assamese, Bengali, Gujarati, Marathi, Tamil, Telugu, Urdu, among others.

Akshay Chaturvedi, founder & CEO, Edtech Startup Leverage Edu,

says, "It is great to see the Union government's focus on education. The FDI part was in waiting for a long time, glad to have it come through—it will really help us make a big leap! Online education programmes, the introduction of new courses that will push students into careers of tomorrow, and of course 'the Indian SAT' for promoting #StudyInIndia—all are in the right direction, and I am personally very excited about us at Leverage Edu helping execute some of these parts."

The National Education Policy for India, whose draft was prepared last year (and the policy has just been announced), aims to transform curriculum to minimise rote learning and encourage holistic development through skills such as critical thinking, digital literacy, scientific temper, communication, multilingualism, problem-solving, and social responsibility. There are plans to optimise resources and avoid unnecessary duplication of content by adding it to an online digital repository, such as the National Digital Library.

On World Youth Skills Day this year, UNICEF India announced its collaboration with SAP India to improve the employability skills of youth in the country for Covid-19 era and beyond it. Besides career counselling, UNICEF will provide digital education and occupational skillset to the underserved young people. The goal is to impact one million people by the end of 2022.

For helping individuals learn these resources and shape the next generation of professionals, higher education institutions have started to incorporate subjects like cloud computing from Amazon Web Services (AWS) and cybersecurity into their curriculum. Standardisation of teaching, learning, and research tools can improve overall skill development.

Recently, Reliance Jio launched its first smart glasses called Jio Glass. It works on mixed reality (MR) and can be powered by both cellular and wireless networks. It supports multiple apps for video conferencing and online collaboration. Besides other purposes, these can also be used for virtual classrooms to teach students online. The company's online education platform called Embibe will be integrated with these smart glasses. Hence, the chances of solving the hurdles seem promising. **EFY**